

A04 CONSTRUCTION POLLUTION MANAGEMENT | P

Intent : Minimize the introduction of construction-related pollutants into indoor air, remediate construction-related indoor air contamination for human health and protect building products from degradation.

Summary : This WELL feature requires projects to protect indoor air quality during building construction and renovation through a combination of strategies, such as duct protection, moisture and dust management, filter replacement, and proper equipment selection.

Issue : Air pollution originating from construction and demolition works can impact the health and quality of life of people working and residing nearby and is associated with increased mortality, due to chronic obstructive pulmonary disease.¹ Increased emissions of PM₁₀ in the vicinity of construction sites are positively correlated with people's suffering from respiratory diseases.² Building construction and renovations are primarily a source of coarse particulate matter, but they are also a source of airborne ultrafine particles.³

Solutions : The indoor air quality levels can be compromised during building construction. Care should be taken during construction to prevent dust intrusion and/or to clear the space of dust, chemical vapors and other debris to avoid significant aggravation of indoor air quality. Air duct protection, moisture and dust management, filter replacement and proper equipment selection are strategies that improve indoor air quality, by limiting the exposure to an intense contamination period.⁴

PART 1 MITIGATE CONSTRUCTION POLLUTION

For All Spaces:

Option 1: Manage construction pollution

For construction occurring after enrollment or the start of subscription, the following requirements are met:

- a. Ducts are maintained per one of the below:
 1. Ducts are sealed and protected from possible contamination during construction.⁵
 2. Ducts are cleaned prior to installing registers, grills and diffusers.
- b. If permanently installed ventilation system is operating during construction, filters must meet the following:
 1. Media filters with an average removal efficiency of $\geq 70\%$ for particles 3-10 μm in size (e.g., MERV 8, M5 or ePM10 70%) are used to filter return air.⁶
 2. All filters are replaced prior to occupancy.⁵
- c. The project implements the following moisture and dust management procedures:
 1. Carpets, acoustical ceiling panels, fabric wall coverings, insulation, upholstery and furnishings and other absorptive materials are stored separately in a designated area protected from moisture damage.⁵
 2. All active areas of work are isolated from other spaces by sealed doorways or windows or through the use of temporary barriers.⁵
 3. Walk-off mats are used at entryways to reduce the transfer of dirt and pollutants.⁵
 4. Saws and similar tools use dust guards or collectors to capture generated dust.⁵

OR

Option 2: No construction activities

The following requirement is met:

- a. The project has not undergone construction activities after enrollment.

REFERENCES

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